

Cloud Computing Architecture

Module 37

Duration: 26:33

Video: https://www.youtube.com/watch?v=-lg-ll_3kAk

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0:00 [Music]

0:23 Hello friends I welcome you all to the video lecture series of cyber security and information security

0:31 in this lecture we will study cloud computing architecture first of all please see the contents which I shall

0:39 cover after covering brief introduction about cloud computing I shall take up essential characteristics of cloud which

0:47 will be followed by three Services offering model and last I shall take up Cloud architecture in the simple terms cloud

0:56 computing means storing and accessing data and programs over the internet from a remote location or computer instead of

1:06 our computer's hard drive it is called remote location and has several properties such as scalability

1:14 elasticity Etc which is significantly different from a simple remote machine the cloud is just a metaphor for

1:23 the internet when we store data on or run a program from the local computer's hard drive that is called local storage

1:33 and Computing for it is to be considered cloud computing we need to access our data or programs over the

1:41 Internet the end result is the same however with an online connection cloud computing can be done anywhere anytime

1:50 and by any device in cloud computing the it and business resources such as servers

1:57 storage Network applications and processes can be dynamically provisioned to the user needs and

2:05 workload in addition while a cloud can provision and support a grid a cloud can also support non-grid environments such

2:14 as three Tiers web architecture running on traditional or web to Applications therefore cloud computing

2:23 is a mechanism of bringing hiring or getting the services of the computing power or infrastructure to an

2:30 organizational or individual level to the extent required and paying only for the consumed

2:37 services in addition one can think to add security while accessing these remote Computing resources as depicted

2:45 in the figure figure shows several cloud computing applications the cloud represents the internet based Computing

2:54 resources and the accessibility is through some secure support of connectivity

3:01 3 minutes, 1 second it is a Computing solution growing in popularity especially among individuals and small and medium sized

3:09 companies thus cloud computing comes into focus and much needed only when we think about what Computing resources and

3:17 Information Technology Solutions are required this need caters to a way to increase capacity or add capabilities on

3:26 the fly without investing in new infrastructure training new personal or lensing new software essential characteristics cloud computing has five essential characteristics

3:33 listeners can note the

3:42 word essential which means that if any of these characterstics is missing then it is not cloud

3:49 computing on demand self-service it has the flexibility of self-provisioning its resources

3:57 according to the demands of the customer without the need of being aided by any human being in network

4:05 access it makes possible for wide range of heterogeneous Network accessibility for mobile clients as well as other

4:13 common computer models broad network access you can access these resources

4:22 from anywhere and you can access them from a browser from a desktop with applications designed to work with them

4:29 or from from a mobile device one of the most popular applications model such as iPhone apps is a mobile application that

4:37 communicates with the cloudbased patent elastic resource pooling the providers Computing

4:46 resources are pooled to serve multiple consumers using a multi-tenant model with different physical and virtual

4:54 resources dynamically assigned and reassigned according to Consumer demand there is a sense of location

5:02 Independence in that the customer generally has no control or knowledge over the exact location of the provided

5:09 resources but may be able to specify the location at a higher level of abstraction
examples of resources

5:17 include storage processing memory and network bandwidth rapid

5:25 elasticity it has the capabilities of managing the pools of resources on demand
basis when need to and free up

5:33 the resources as situation demands location independent resource pooling the
resources for the customers are

5:42 aggregated in multi-tenant model irrespective of the geolocation of the consumers
the customers have no idea

5:51 about the location of the resources he she were provision from except some
abstract high level of information like

5:59 countries and regions to the consumer the capabilities available for provisioning
often appear to be

6:06 unlimited and can be purchased in any quantity at any time Meed

6:14 service cloud computing has a metering capability which controls monitors and
reports the use of

6:22 resources this results in audited reports for both the provider and the consumer
showing exec

6:30 how much hard disk was used which Services were accessed and for how many
hours the implication for companies that

6:38 use cloud computing means they pay only for what they use which is very coste
efficient for business

6:45 operations resource uses can be monitored controlled and reported providing
transparency for both the

6:53 provider and consumer of the utilized service cloud depl deployment

7:01 7 minutes, 1 second models deployment models describe the ways with which the
cloud services can be deployed or made available to its

7:09 customers depending on the organizational structure and the provisioning location
Cloud that is internet based

7:18 Computing resources that is the locations where data and services are acquired
and provision to its customers

7:25 can take various forms for deployment models are public Cloud private Cloud

7:33 Community cloud and hybrid Cloud private

7:39 Cloud it is also known as internal Cloud the platform for cloud computing is

7:46 implemented on the cloud-based secure environment that is safeguarded by a
firewall which is under the governance

7:54 of the IT department that belongs to the particular corporate private Cloud as it
permits only the

8:02 authorized users has direct control over their data what exactly constitutes a

private Cloud it is difficult to Define

- 8:11 because when it's classified according to the services there are significant variations whether the physical
- 8:18 computers are hosted internally or externally they provide the resources from a distinct pool to a private cloud
- 8:28 services business reses that have Dynamic needs assignments which are mission critical Security Alarms
- 8:36 management demands and uptime requirements are better suited to adopt private Cloud obstacles with regards to
- 8:44 security can be evaded in a private Cloud that in case of natural disaster and internal Data Theft the private
- 8:51 Cloud may be prone to vulnerabilities public Cloud public cloud is a type of of cloud
- 9:00 hosting in which the cloud services are delivered over a network which is open for public usage this model is a true
- 9:09 representation of cloud hosting the customers do not have any distinguished ability and control over the location of
- 9:17 the infrastructure from the technical Viewpoint there may be slight or no difference between private and public
- 9:24 Cloud's structural design except in the level of security offered for various Services given to the public Cloud
- 9:32 subscribers by the cloud hosting providers public cloud is better suited for business requirements which require
- 9:40 managing the load host application that is saas based and manage applications
- 9:47 that many users consume due to the decreased capital overheads and operational cost this model is
- 9:55 economical the dealer May provide the service free or in the form of the license policy like pay per user the
- 10:04 cost is shared by all the users so public Cloud profits the customers more by achieving economies of scale public
- 10:13 Cloud facilities may be availed free example a public cloud is
- 10:20 Google Community Cloud it is a type of cloud hosting in which the setup is mutually shared between many
- 10:28 organizations that belong to a particular community that is Banks and trading
- 10:35 firms it is a multi setup that is shared among several organizations that belong to a specific group which has a similar
- 10:44 Computing apprehensions the community members generally share similar privacy performance and security concerns the
- 10:53 main intention of these communities is to achieve their business related objectives a Community Cloud may be

11:00 internally managed or it can be managed by a third party provider it can be hosted externally or internally the cost

11:09 is shared by the specific organizations within the community hence Community cloud has cost saving

11:16 capacity organizations have understood that cloud hosting has a lot of potential so selection of the right type

11:24 of cloud hosting is needed thus you need to know your business and analyze the demands once the appropriate type of

11:33 cloud hosting is selected you can achieve your business related goals more easily you can channelize all your

11:40 efforts to take those strategic steps that will help your business to succeed hybrid Cloud it is a type of

11:49 cloud computing which is integrated it can be an arrangement of two or more Cloud servers that is

11:57 private public or Comm com Cloud that is bound together but remain individual entities benefits of the multiple

12:06 deployment models are available in hybrid Cloud hosting a hybrid Cloud can cross isolation and overcome boundaries

12:14 by the provider hence it cannot be simply categorized into public private or Community Cloud it permits the user

12:22 to increase the capacity or the capability by aggregation assimilation or customization with another Cloud

12:30 package or service in a hybrid Cloud the resources are managed and provided either inhouse

12:37 or by external providers it is an adaptation among two Platforms in which the workload exchanges between the private cloud and

12:46 the public Cloud as per the need and demand resources that are non critical like development and test workloads can

12:54 be housed in the public Cloud that belongs to a third party provider while the workload set are critical or

13:02 sensitive must be housed internally consider an e-commerce website which is hosted on a private

13:09 Cloud that gives security and scalability since security is not a prime concern for those brocher site it

13:19 is hosted on a public Cloud which is more economical as compared to a private Cloud businesses that have more focus on

13:28 security and demand for their unique presence can Implement hybrid Cloud as an effective business strategy when

13:36 facing demand spikes the additional resources that are required by a particular application can be accessed from the public

13:44 Cloud this is termed as Cloud bursting and is available with the hybrid
13:51 Cloud organizations can use the hybrid cloud model for processing big data on a
13:58 private cloud it can retain sales business and various data and can initiate
analytical queries over the
14:05 public Cloud as the public cloud is effective to meet the demand spikes hybrid
cloud hostings is enabled
14:13 with features like scalability flexibility and security if one is ready to overlook a few
challenges like
14:20 application program interface in compatibility network connectivity issues and
capital expenditures then
14:28 hybrid Cloud would be an appropriate option three service offering models the
three kinds of services with which the
14:37 cloud-based Computing resources are available to end customers are software as a
service
14:45 saas platform as service PaaS and infrastructure as a service
14:54 IAS it is also known as the service platform in infrastructure SBI model of
15:01 15 minutes, 1 second the cloud and it is shown in the figure SaaS is a software
distribution model in
15:10 which applications are hosted by a vendor or service provider and made available
to customers over a
15:18 network PaaS is a paradigm for delivering operating systems and Associated
15:26 Services example computer AED software engineering C case tools integrated
15:33 development environments IDEs for developing software Solutions over the internet
without downloads or
15:43 installation IAS involves Outsourcing the equipment used to support operations
including
15:51 storage Hardware servers and networking components Cloud is a
16:00 the capability provided to the consumer is to use the provider's applications
running on a cloud infrastructure
16:09 including network servers operating systems storage and even individual
application capabilities with the
16:17 possible exception of limited user specific application configuration settings the
applications are accessible
16:24 from various client devices through either a Thin Client interface such as web
browser or a program
16:32 interface the consumer does not manage or control the underlying Cloud
infrastructure typical applications
16:39 offered as a service include customer relationship management CRM business
intelligence analytics and

16:48 online accounting software Cloud paaS the capability provided to Consumer
16:57 is to deploy onto the Cloud infrastructure the consumer created or acquired applications created using
17:04 programming languages libraries services and tools supported by the provider the consumer does not manage or
17:14 control the underlying Cloud infrastructure but has control over the deployed applications and possibly configuration settings for the
17:22 application hosting environment in other words it is a packaged and ready to run development or
17:31 operating framework the PaaS window provides the networks servers storage
17:39 and manages the levels of scalability and maintenance the client typically pays for services used examples of PaaS
17:49 providers include Google App Engine and Microsoft Azure
17:55 Services Cloud IaaS the capability provided to the consumer
18:02 is to provision processing storage networks and other fundamental Computing resources on a pay per use basis where he
18:12 or she is able to deploy and run software which can include operating systems and applications the consumer does not
18:20 manage or control the underlying Cloud infrastructure but has control over the operating systems storage and deployed
18:28 applications and possibly limited control of select networking components the service provider owns the
18:36 equipment and is responsible for housing cooling operation and maintenance Amazon Web Services AWS is a
18:46 popular example of a large IaaS provider the major difference between
18:53 PaaS and IaaS is the amount of control that you users have in essence PaaS allows
19:03 vendors to manage everything while IaaS requires more management from the customer
19:11 site generally speaking organizations that already have a software package or application for a specific purpose and
19:19 want to install and run it in on the cloud should opt to use
19:25 IaaS instead of PaaS so now we come over to the cloud
19:33 architecture the cloud has a high level view of an architecture that describes its working mechanism it includes the
19:41 dependencies on which it works and the components that work over it the cloud is a recent technology that is
19:49 completely dependent on the internet for its functioning figure depicts the architecture the cloud architecture can
19:57 be divided into four layers based on the access of the cloud by the user they are

20:04 as follows layer one that is user client layer this layer is the lowest layer in
20:11 the cloud architecture all the users or client belong to this layer this is the place
where the client user initiates
20:19 the connections to the cloud the client can be any device such as a Thin Client
20:26 thick client or mobile or any handheld device that would support basic
functionalities to access a web
20:34 application the 10 client here refers to a device that is completely dependent on
some other system for its complete
20:42 functionality in simple terms they have very low processing capability similarly
thick clients are
20:49 General computers that have adequate processing capability they have sufficient
capability for independent
20:56 work usually a cloud Cloud application can be accessed in the same way as web
application but internally the
21:04 properties of cloud applications are significantly different thus this layer consists of
a client
21:12 devices Layer Two that is Network layer this layer allows the users to connect to
the cloud the whole Cloud
21:20 infrastructure is dependent on this connection where the services are offered to
customers this is primarily the internet in the case of a public
21:28 Cloud the public Cloud usually exists in the specific location and the user would not
know the location as it is abstract
21:38 and the public Cloud can be existed all over the world in the case of a private Cloud
the connectivity may be provided
21:46 by a local area network lamb even in this case the cloud completely depends on the
network that is used usually when
21:55 accessing the public or private Cloud the US users require minimum bandwidth
which is sometimes defined by the cloud
22:03 providers this layer does not come under the perview of a service level agreements
22:10 slas slas do not take into account the internet connection between the user and
Cloud for quality of service that is
22:20 Qs layer three that is cloud management layer this layer consists of softwares
22:27 that are used in managing the cloud these softwares can be Cloud operating system
OS which acts as an interface
22:35 between the data center and the user or a management software that allows
managing resources these softwares usually allow
22:44 Resource Management that is scheduling provisioning Etc optimization server
consolidation storage workload
22:52 consolidation and internal Cloud governance this layer comes under the view of
slas that is the operations

23:01 23 minutes, 1 second taking place in this layer would affect the slas that are being decided upon between the users and service

23:09 providers any delay in processing or any discrepancy in service provisioning may lead to an SLA violation as per rules

23:18 any SLA violation would result in penalty to be given by the service provider these slas are for both private

23:27 and public clouds popular service providers are Amazon web services AWS

23:34 and Microsoft azure for public Cloud similarly open stack allow private Cloud

23:42 creation deployment and management layer four Hardware resource

23:49 layer layer four consists of Provisions for actual Hardware sources usually in the case of a public Cloud a data center

23:57 is used in the back end similarly in a private Cloud it can be data center which is a huge collection of Hardware

24:05 sources interconnected to each other that is present in a specific location or a high configuration system this

24:12 layer comes under the purview of slas this is the most important layer that governs the slas this layer affects

24:21 the slas most in the case of data centers whenever a user accesses the cloud it should be available to the

24:29 users as quickly as possible and should be within the time that is defined by the slas if there is any discrepancy in

24:37 provisioning the resources or applications the service provider has to pay the penalty hence the data center

24:45 consist of high speed network connection and a highly efficient algorithm to transfer the data from the data center to the manager there can be

24:55 a number of data centers for a cloud and similarly a number of clouds can share a data

25:01 25 minutes, 1 second center thus this is the architecture of a cloud there can be little loose isolation between layer three and layer

25:10 four depending on the way the cloud is deployed so friends this was all about cloud computing and its

25:19 architecture hope the concepts explained in this lecture were understandable and helpful hope to see you in the next

25:27 lecture till then goodbye goodbye thank you

25:31 [Music]